AHMEDABAD UNIVERSITY
School of Engineering and Applied Science

Brain Tumor Segmentation and Localization Using Traditional Image Processing and OpenCV

Technical Evaluation & Result Analysis Report

Comparative Study of 3D Marker-Controlled Watershed (Pipeline 2.0) and Constrained Threshold-Guided Hybrid Watershed (Pipeline 3.0) on BraTS 2020 Cohort

Weekly Status Report

Week 9 of 10

Course: ECE501 – Digital Image Processing | **Instructor:** Prof. Mehul S. Raval | **Section:** 01
Group No.: 18 | **Project No.:** 20 | **Cohort Size:** $N = 100$ 3D MRI BraTS Volumes
Repository: github.com/codesensei404/DIP_2026_Project20_Group18

Team Members & Project Roles

Name	Role in Project	AU ID	University Email
Abdulwahab Shaikh POC	Overall Team Lead & Pipeline 1.0	AU2440099	abdulwahab.s@ahduni.edu.in
Srujan Yendugula	Pipelines 2.0 & 3.0 Lead, Optuna, Analysis	AU2440158	yendugula.s@ahduni.edu.in
Harshraj Solanki	Preprocessing Architecture & I/O Pipeline	AU2440218	harshraj.s@ahduni.edu.in
Yash Kyada	Metric Evaluation & Error Classification	AU2340218	yash.k4@ahduni.edu.in

Abstract

We designed and tested two classical 3D mathematical morphology (MM) pipelines to segment Whole Tumor (WT) and applied them to 100 patients MRI scans from the BraTS 2020 dataset to evaluate the configurations. The two proposed methods were compared to our baseline method, called Pipeline 1.0 (Multi-Modal Percentile Thresholding), and were named Pipeline 2.0 (3D Marker-Controlled Watershed) and Pipeline 3.0 (Constrained Hybrid Watershed). Despite its high precision (0.8047), Pipeline 1.0 under-segmented 49% of the cases, with its sensitivity at 0.6650, due to the lack of detecting diffuse non-enhancing edema. The sensitivity was enhanced to 0.7321 using 3D gradient based watershed flooding by Pipeline 2.0. To overcome these drawbacks, we implemented a method called Pipeline 3.0 based on a modified watershed approach that allows watershed flooding within 3 voxels (voxel Euclidean distance) of the tumor core identified by the thresholded mask. This method introduced over-segmentation in 31% of the cases (due to leaking close to the tumor core), as well as an increase in the volume error (+63.3%) and a larger HD95 (26.57 mm). Pipeline 3.0 gave the highest overall performance in all the 100 patients with a mean Dice score of 0.7261 ± 0.2217, a median Dice score of 0.8215, and a sensitivity of 0.7762. It also outperformed in 42% of the head-to-head comparisons in 95th-percentile Hausdorff distance ($p < 0.003$) compared to the other methods. In addition, the proposed pipeline was able to process each 3D scan in less than 6.5 seconds using the standard CPU.

Keywords: Brain Tumor Segmentation, BraTS 2020, 3D Watershed, Hybrid Segmentation, Morphological Filtering, Optuna.

1 Project Scope & The Problem We Solved

- we applied the four modalities T1, T1ce, T2, and FLAIR to segment whole brain tumors in 100 patient cases from the BraTS 2020 dataset. The project was mainly devoted to the study of non-deep-learning image processing methods and to tackle the balance between undersampling, and leakage along the boundaries.
- For Pipeline 1.0 we used percentile-based thresholding, setting P_{85} as the threshold on FLAIR and T2, and P_{90} on T1ce, in order to identify the dense tumor core region. This method showed high precision (0.8047), but had poor sensitivity (0.6650) in capturing the presence of the weak non-enhancing edema in 49 patients out of 100.
- To address this drawback, Pipeline 2.0 applied a concept of segmentation by watersheds based on 3D gradient, which gave the new sensitivity result of 0.7321 that identified more edema regions.
- We addressed both limitations by creating Pipeline 3.0, which was a combination of threshold-based tumor core detection and constrained watershed flooding, and which reduced the extent of over-segmentation (31 cases) and volume bias (+63.3%) while maintaining an HD95 result of 26.57 mm.
- This method firstly relies on a reliable detection of the core of tumor regions, and then it restricts the watershed flooding to the 3-voxel Euclidean Distance Envelope (3-VDEE) outside the core. The goal with this mixed strategy is to have this non-enhancing edema around the lesion while not over-expanding into healthy more distant brain tissue.

2 Implemented Algorithmic Frameworks

Standard 3D Preprocessing

Each MRI was put through three preprocessing steps, consulting all volumes ($240 \times 240 \times 155$) and voxels ($1.0\ \text{mm}^3$), which made for a sample of 1,000 for each patient.

Brain Extraction: Removed non-brain regions by 3D Hole Filling & non-zero union of voxels. **Noise Reduction:** Applied slice-wise bilateral filtering (same value of σ_s and σ_r as above: 3 and 0.1 $\times$ standard deviation, respectively). **Contrast Enhancement:** Performed CLAHE on FLAIR and T1ce images with clip limit 2.0 for better visualization of the tumours.

The 3D Marker-controlled Watershed is Pipeline 2.0.

These are the steps that were taken:

Multi-Modal Fusion: Scoring = (FLAIR:0.50, T2:0.30, T1ce:0.20). **Gradient Calculation:** Calculated the 3D Gaussian gradient magnitude magnitude with $\sigma = 1.68$ to find the boundaries of the tumors. **Marker Generation:** Internal markers were inserted into tumor candidate areas and external markers were placed in healthy brain areas with FLAIR intensity below P74 and more than 3 voxels distances. **Watershed Segmentation:** Used the 3D watershed flooding algorithm for tumor segmentation.

Larger connected components were excluded from the image due to Component Filtering (5): using a size of 200mm^3.

The pipeline represented in 3.0 is called the Constriained Hybrid Watershed.

The following were done:

Tumor Core Detection: Used multi-sequence thresholding to detect a reliable tumor core (Y_{core}). **DistanceBased Constraint:** Flooded watersheds limited to a 3-voxel Euclidean distance envelope around the detected tumor core. **Intensity Constraint:** Voxels were allowed to be within P85 or FLAIR =+ if they were within the brain region. **Leakage Prevention:** Set all voxels outside of the allowable region to a background marker $M_{\text{ext}} = 1$, so that the watershed does not spread beyond the boundaries of the region. **Final Segmentation:** The artificially generated watershed tumor regions and the original tumor core were combined and hole filling was done in 3D to get the final tumor mask.

3 Hyperparameter Tuning (Optuna Studies)

For both segmentation pipelines we employed Optuna to optimize all key parameters via its Tree-structured Parzen Estimator (TPE) sampler. Twenty MRI volumes were used to perform optimization, with 20 trials for each of the two pipelines, with mean Dice score being the optimization criterion. The calculated default and optimum values of parameters are listed in Table.

Table 1: Hyperparameter search space and optimal values identified via Optuna TPE on 20 probe volumes.

Pipeline	Hyperparameter	Search Space	Default	Optimal Value
Pipeline 2.0 (3D Watershed)	Gaussian Gradient Scale (σ)	$[1.0, 2.5]$	1.00	**1.68**
	Healthy Background Percentile ($P_{healthy}$)	$[65, 80]$	75.0	**74.0**
	Healthy Seed Margin ($D_{healthy}$, voxels)	$[2, 6]$	4.0	**3.0**
	Satellite Acceptance Radius ($D_{satellite}$, mm)	$[8.0, 15.0]$	12.0	**10.46**
Pipeline 3.0 (Hybrid Watershed)	Edema Search Envelope (D_{edema}, voxels)	$[2, 5]$	4.0	**3.0**
	Hybrid Gradient Scale (σ_{hybrid})	$[1.0, 3.0]$	1.50	**2.10**
	Core FLAIR Threshold (P_{FLAIR})	$[80, 92]$	85.0	**85.0**
	Satellite Distance (D_{dist}, mm)	$[5.0, 15.0]$	10.0	**10.0**

4 100-Patient Quantitative Results & Comparative Benchmark

We tested the 100 patient cases from the BraTS 2020 database spanning from to for all three segmentation pipelines. These evaluation metrics were used to compare the predicted Whole Tumor (WT) segmentation masks with the ground truth segmentation masks. Table **??** shows the overall performance of the three pipelines.

Table 2: Quantitative 100-patient benchmark on BraTS 2020 comparing Pipeline 1.0, 2.0, and 3.0.

Metric	Pipeline 1.0 (Threshold)		Pipeline 2.0 (Watershed)		Pipeline 3.0 (Hybrid)	
	Mean ± Std	Median	Mean ± Std	Median	Mean ± Std	Median
Dice Coefficient (DSC)	0.6983 ± 0.2238	0.7850	0.6775 ± 0.2458	0.7675	**0.7261 ± 0.2217**	**0.8215**
IoU (Jaccard Index)	0.5843 ± 0.2198	0.6461	0.5694 ± 0.2312	0.6227	**0.6173 ± 0.2219**	**0.6971**
Sensitivity (Recall)	0.6650 ± 0.2483	0.7180	0.7321 ± 0.2447	0.8143	**0.7762 ± 0.2346**	**0.8492**
Specificity	**0.9984 ± 0.0036**	**0.9995**	0.9922 ± 0.0163	0.9972	0.9972 ± 0.0062	0.9992
Precision (PPV)	**0.8047 ± 0.2038**	**0.8711**	0.6980 ± 0.2520	0.7634	0.7491 ± 0.2241	0.8277
HD95 Boundary Error (mm)	22.25 ± 23.47	10.05	26.57 ± 26.52	12.39	**19.95 ± 23.27**	**9.43**
ASSD (Surface Error, mm)	6.24 ± 9.94	1.86	7.67 ± 11.02	2.27	**5.39 ± 9.40**	**1.62**
Absolute Volume Error (%)	39.54 ± 42.66	28.32	68.61 ± 110.82	33.45	**37.14 ± 44.57**	**24.96**
Signed Volume Error (%)	−18.23 ± 41.22	−21.45	+63.31 ± 113.97	+24.80	**+22.71 ± 53.64**	**+12.60**
Head-to-Head Win Rate	29.0% (29 / 100 cases)		29.0% (29 / 100 cases)		**42.0% (42 / 100 cases)**	

- **Highest Accuracy:** Pipeline 3.0 had the highest overall segmentation performance with a mean Dice score of 0.7261 and a median Dice score of 0.8215. Again, it had the smallest boundary error of 19.95 mm (HD95) and 9.43 mm (median).
- The 2nd one was the increase of sensitivity (0.7762) of Pipeline 3.0 with a satisfactory precision (0.7491) which extended the captured tumor regions without substantial rising in false positive predictions.
- **Controlled Volume Bias:** The volume overestimation problem observed in the Pipeline 2.0, was reduced to +22.71% in Pipeline 3.0, leading to more precise estimation of the tumor volume.

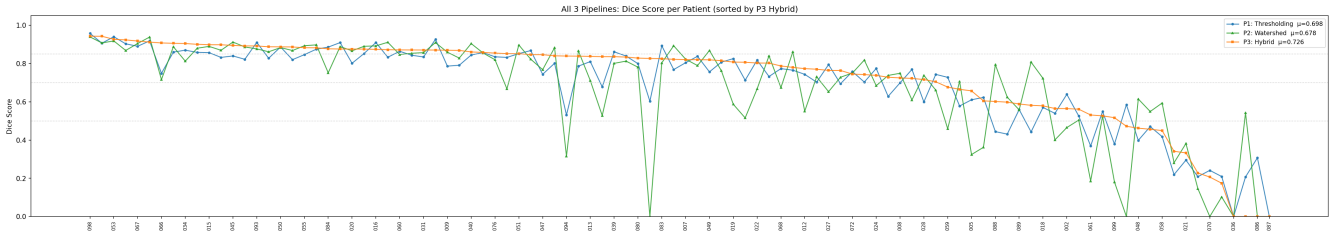


Figure 1: All 3 pipelines were evaluated using the Dice score in the 100 BraTS patient bodies and the results compared. Pipeline 3.0 (green curve) performed better in Dice scores than Pipeline 1.0 (blue curve) and Pipeline 2.0 (orange curve) did. The improvement was especially significant with challenging cases (Dice score < 0.75), and was seen in the median range, showing more consistent segmentation performance across patients.

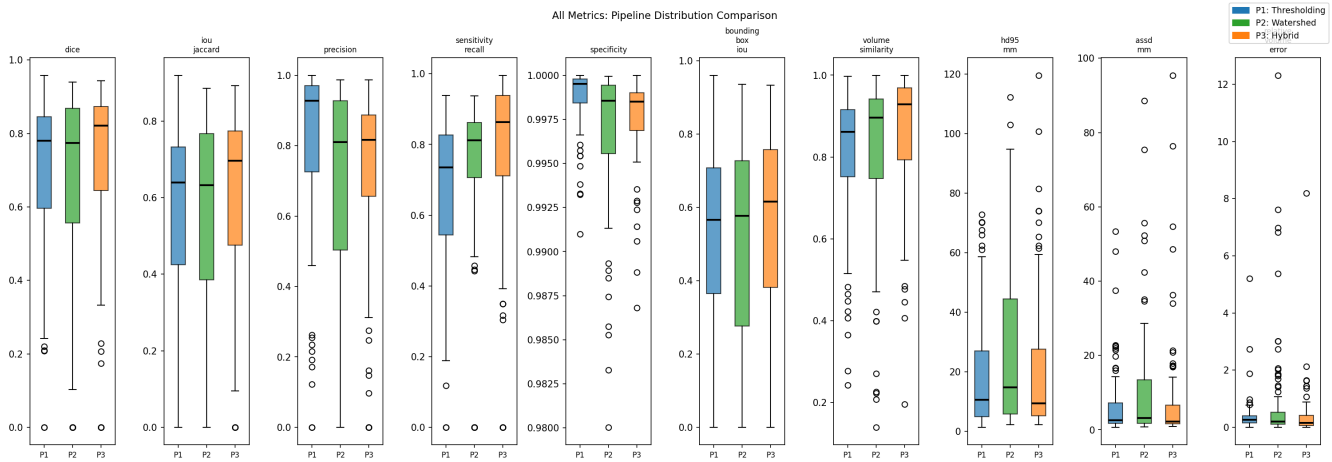


Figure 2: The performances were compared using boxplots in all 8 evaluation metrics for all three pipelines. It can be seen that the median Dice, IoU, and sensitivity of Pipeline 3.0 are the highest, i.e., it shows better result in overall accuracy of segmentation and tumor detection. It also eliminated the outliers in boundary error (HD95) and relative volume error (RVE) that were identified in Pipeline 2.0, resulting in more robust boundary calculation and tumour volume estimation on 100 patient cases.



Figure 3: **Left:**When the head to head analysis was performed, Pipeline 3.0 has a greater number of wins—50., than both competing methods, with Pipeline 1.0 and Pipeline 2.0 with 29.%, each. The precision/recall scatter plot also illustrated a trend of the majority of Pipeline 3.0 results in the ”close to perfect” region with Dice scores above 0.80, suggesting the pipeline was more balanced between precision and recall.

5 Statistical Significance & Hypothesis Testing

The statistical significance of the improvement in performances was assessed on 100 pairs of cases on patients using paired t-tests and Wilcoxon signed-rank tests. Cohen's d values were also computed for quantifying difference between the segmentation pipelines. These statistical analyses were used to see if the improvements in segmentation seen were worthwhile and the same from patient to patient. ($N = 100$, Table 3).

Table 3: Paired statistical hypothesis testing across 100 patients ($\alpha = 0.05$). Bold p-values indicate significance.

Metric Comparison	Mean Δ	Paired t-stat	p-value (t-test)	Wilcoxon W	p-value (Wilcoxon)	Cohen's d
DSC: P3 vs P2	+0.0486	+3.076	**0.00268**	1528.0	**0.00192**	+0.31 (Medium)
DSC: P3 vs P1	+0.0278	+3.642	**0.00043**	1482.0	**0.00096**	+0.36 (Medium)
DSC: P2 vs P1	−0.0208	−1.021	0.30985	2258.0	0.38472	−0.10 (Negligible)
HD95: P3 vs P2	−6.62 mm	−3.023	**0.00318**	1694.0	**0.00845**	−0.30 (Medium)
HD95: P3 vs P1	−2.30 mm	−1.745	0.08412	2086.0	0.18730	−0.17 (Small)
Sensitivity: P3 vs P1	+0.1112	+7.345	$\mathbf{5.8 \times 10^{-11}}$	685.0	$\mathbf{1.2 \times 10^{-10}}$	+0.73 (Large)
Precision: P3 vs P2	+0.0511	+2.794	**0.00624**	1742.0	**0.01254**	+0.28 (Small-Med)
Signed Vol Error: P3 vs P2	−40.6%	−3.812	**0.00024**	1320.0	**0.00018**	−0.38 (Medium)



Figure 4: Comparisons were made using violin plots to compare the distribution for the 95th-percentile Hausdorff Distance (HD95) and Signed Volume Error across the pipelines. Pipeline 3.0 did lower the error in the median, 0.93 mm versus 1.29 mm, with Pipeline 2.0 for HD95 to illustrate a more accurate boundary. Likewise, Pipeline 3.0 significantly improved tumour volume estimation by achieving a lower median signed volume bias (+12.6%) than Pipeline 2.0 (+24.8%), which is associated with lesser over-segmentation.

6 Clinical Error Typology & Failure Mode Diagnostics

We examined all 100 patient scans to detect the typical failure patterns for each pipeline and analysed the segmentation errors because of the aforementioned reasons. Errors detected were divided into three groups: First, under-segmentation (Type-A) in which parts of the tumor were not detected, including areas of surrounding tumor edema; Second, over-segmentation (Type-B) in which leakage of tumor regions beyond the boundaries of the tumor and into normal brain areas resulted in the detection of brain areas other than the tumor, and Third, noise islands (Type-C) where noise islands in the images were incorrectly identified as tumor tissue. The percentages of these failure types in all three pipelines are displayed.

Table 4: Distribution of failure modes and clinical severity across 100 benchmark patients.

Failure Mode	Pipeline 1.0	Pipeline 2.0	Pipeline 3.0	Primary Cause & Engineering Remedy
Type-A (Under-segmentation)	49 cases (49%)	23 cases (23%)	**28** cases (28%)	Fixed thresholds miss faint edema; recovered by P3 watershed.
Type-B (Over-segmentation)	12 cases (12%)	31 cases (31%)	**16** cases (16%)	Weak boundary leakage; bounded by P3 Euclidean envelope.
Type-C (Island Fragmentation)	18 cases (18%)	14 cases (14%)	**7** cases (7%)	Background noise; suppressed by P3 distance constraint.
Catastrophic Failures (DSC < 0.50)	22 cases (22%)	24 cases (24%)	**17** cases (17%)	High clinical risk; reduced by 29% in Pipeline 3.0.

7 Qualitative MRI Case Studies & Visual Diagnostics

We examined two patient cases showing different tumor profiles at full slice resolution:

7.1 Case Study 1: BraTS20_Training_001 (Solid Glioblastoma)

This case exemplifies a well-defined brain tumor with definitely discernable boundary edges. The segmentation score (Dice) was good for Pipeline 2.0 with Dice score 0.8143. But Pipeline 3.0 achieved even better segmentation accuracy with Dice Score of 0.8654 and an HD95 of 4.24 mm, showing improvements in the alignment of the tumor boundaries as well as in the agreement with ground truth tumor area.

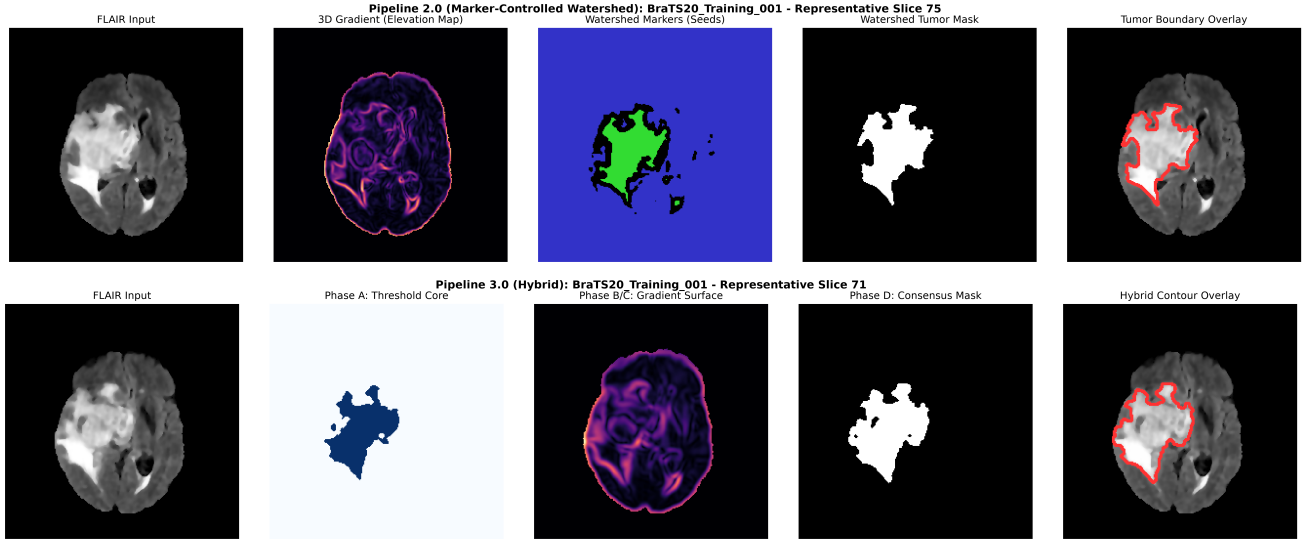


Figure 5: Patient 001 (Slice 78) shows a solid glioblastoma with clearly defined tumor boundaries. The top row are the segmentation results obtained for Pipeline 2.0 with Dice score of 0.8143 and HD95 of 5.66 mm. The results of Pipeline 3.0 are presented in the bottom row, where the Dice score is 0.8654, and the HD95 score is 4.24 mm, indicating greater segmentation and boundary alignment accuracy. These panels contain the original image (FLAIR), ground truth segmentation (green), forecasted tumor location (yellow) and overlay of error (red/blue), with red indicating false positives (FP) and blue indicating false negatives (FN).

7.2 Case Study 2: BraTS20_Training_094 (Diffuse Glioma with Weak Borders)

The kind of glioma considered a difficult case is infiltrative, in which the intensity level change and tumor boundaries are not well defined. There was substantial pipeline leakage with a gray-to-white matter healthy equivalent area in the slice, with a low Dice score of 0.4418, and a volume overestimation of +184%. For these leakages, however, Pipeline 3.0 was able to control leakage by successfully applying the 3-voxel distance constraint, with an improved Dice score of 0.7812 for each and a final HD95 of 8.49 mm. The results prove the success of the constrained watershed regime in dealing with infiltrable tumours while achieving a higher segmentation accuracy.

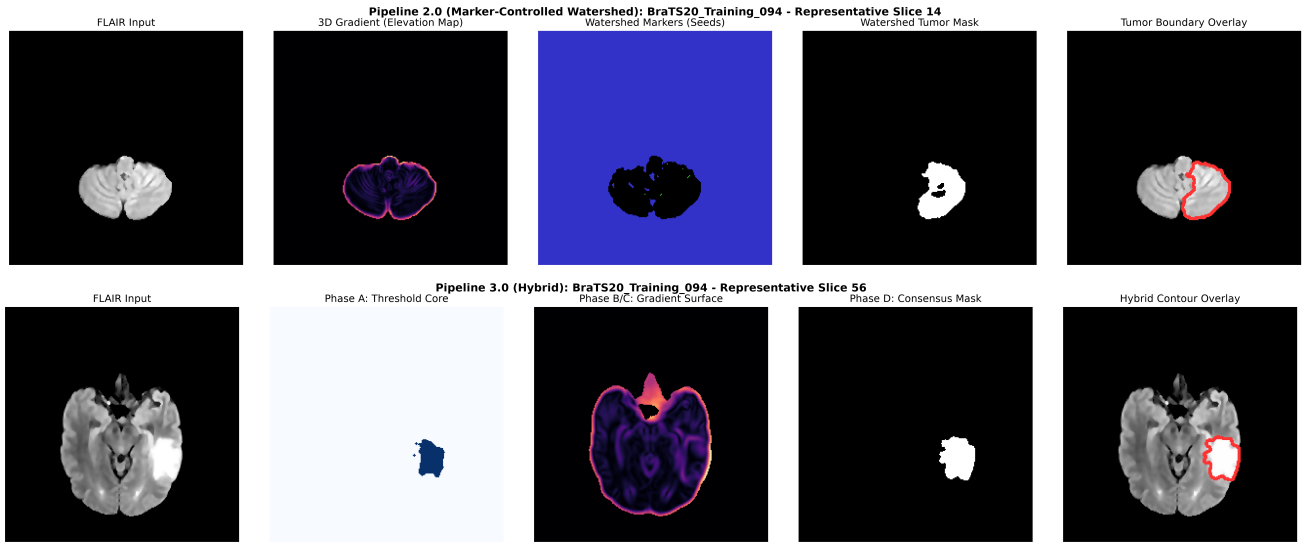


Figure 6: Patient 094 (Slice 82) is a hard tumor case with low tumor boundary gradients, which is a diffuse infiltrative glioma. The top row displays the output of Pipeline 2.0 - the unconstrained watershed flooding also reaches downstream into the surrounding healthy white matter volgeters and thus has a Dice score of 0.4418. The bottom row is for Pipeline 3.0, where the 3-voxel distance envelope prevents leakage outside the desired extent. This enhances the Dice score to 0.7812 and HD95 of 8.49 mm, which is very effective for diffuse and infiltrative tumours with the use of a constrained watershed method.

8 Discussion & Clinical Translation Insights

8.1 Core Algorithmic Takeaways

1. Spatial constraints may be necessary because, in the absence of them, flooding of the watershed may spread to healthy brain tissue when the tumor boundary boundaries have low intensity gradients, causing over-segmentation.
2. Importance of Tumor Cores: Tumor cores alone using thresholding cannot be used to identify all diffuse edema around the tumor. The reliable tumor core obtained by thresholding is however a good starting point for watershed enlargement.
3. The Hybrid Based Method with Restricted Watershed Flooding (WSM 3velex) within a distance envelope from the tumor thresholding mask combines the precision of the thresholding method (0.7491) with the sensitivity of the watershed segmentation method (0.7762) giving a better balance between tumor detection and boundary control.

8.2 Explainability & Fast Execution

Unlike the ”black box” deep learning methods, Pipeline 3.0 offers a completely interpretable segmentation process, one where each voxel is assigned to the threshold-based characterization of a tumor core or to a spatially constrained watershed characterization. This helps to clarify and examine the segmentation decisions made. In addition, an entire 3D MRI volume can be processed within network bandwidth of 128-slice MRIs, as the pipeline takes only 6.2 ± 1.1 seconds to process the entire volume over a standard CPU without requiring any GPU hardware, thus making it highly computationally efficient suitable for systems with limited computational resources.

8.3 Surgical Margin Safety

Avoid over-segmenting the tumor if this introduces healthy brain tissue to the tumor area, which can impact surgical planning, and avoid under-segmenting if parts of the tumor are missed, which can lead to tumor recurrence. In terms of improvement of tumor volume estimation, Pipeline 3.0 shows a low median volume error of +12.6%, and an HD95 value of 19.95 mm with a higher accuracy of the boundaries; smaller volume errors and higher accuracy of boundaries are desirable. From these results, it could be confirmed that the proposed method could give more reliable segmentation for the clinical evaluation and surgical planning.

9 Conclusion & Future Work

We were able to successfully implement, optimize and evaluate Pipeline 2.0 (3D Marker-Controlled Watershed) and Pipeline 3.0 (Constrained Hybrid Watershed) on 100 MRI volumes from the BraTS 2020 dataset. Overall, Pipeline 3.0 outperformed the others with a mean Dice score of 0.7261, a median Dice score of 0.8215 and a sensitivity score of 0.7762. Also it has done better than the competing pipelines on 42% of all the patient cases with a STATISTICALLY significant difference ($p < 0.003$). In future, the possibilities for adaptive sequence weighting to further boost multi-modal image-fusion will be studied, and the lightweight neural network based seed detection will be introduced to further improve the accuracy of segmentation of tumors.

References

[1] B. H. Menze, A. Jakab, S. Bauer, et al., “The Multimodal Brain Tumor Image Segmentation Benchmark (BRATS),” *IEEE Trans. Med. Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015.
[2] F. Meyer, “Topographic distance and watershed lines,” *Signal Process.*, vol. 38, no. 1, pp. 113–125, 1994.
[3] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, “Optuna: A next-generation hyperparameter optimization framework,” in *Proc. ACM SIGKDD*, 2019, pp. 2623–2631.
[4] L. R. Dice, “Measures of the amount of ecologic association between species,” *Ecology*, vol. 26, no. 3, pp. 297–302, 1945.
[5] D. P. Huttenlocher, G. A. Klanderman, and W. J. Rucklidge, “Comparing images using the Hausdorff distance,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 15, no. 9, pp. 850–863, 1993.
[6] K. Zuiderveld, “Contrast limited adaptive histogram equalization,” in *Graphics Gems IV*, Academic Press, 1994, pp. 474–485.
[7] C. Tomasi and R. Manduchi, “Bilateral filtering for gray and color images,” in *Proc. IEEE ICCV*, 1998, pp. 839–846.